

Modeling Work

1. Classification models (churn prediction)

customer_churn_predictions.csv (now the customer_churn_predictions table) has a churn_probability and churn_risk column for every one of the 7,043 customers — e.g. customer 7590-VHVEG scored 0.375 (Low Risk), customer 5575-GNVDE scored 0.042 (Low Risk). This means someone on the team already trained a classifier (per the doc's plan: Logistic Regression, Random Forest, and/or XGBoost), picked a winner, and ran it on the full dataset to generate these probabilities. The Week 2 timeline item "train classification models... evaluate using precision, recall, F1-score" is done.

2. Feature importance (from the tree-based model itself)

feature_importance.csv — top drivers the model found: InternetService_Fiber optic (0.242), Contract_Two year (0.223), Contract_One year (0.210). This is the built-in .feature_importances_ output from whichever tree-based model (Random Forest or XGBoost) won.

3. SHAP explainability

shap_feature_importance.csv + shap_summary.png — SHAP's own ranking, which is different from #2: Contract_Two year (0.557) and tenure (0.505) top the SHAP list instead. This is the Week 2, Day 7 task — "implement SHAP values to explain feature importance for business stakeholders" — already done, and the PNG is literally the summary plot meant for the stakeholder presentation.

4. LTV regression

customer_ltv_segments.csv has a predicted_ltv value per customer plus a customer_segment label, split into 4 buckets: Low Value–Low Risk, High Value–Low Risk, Low Value–High Risk, High Value–High Risk. This is the Week 3, Day 1–3 task — "develop regression models to forecast the expected lifetime revenue of active customers" — already done.

5. A derived output: priority retention list

priority_retention_customers.csv — 166 customers, exactly the ones flagged High Value – High Risk. This isn't a separate model; it's a filter on top of #1 and #4 combined (this is why I set it up as a database view rather than its own stored table — it's a query, not new data).

So in short: **the ML work for Weeks 2 and most of Week 3 is done.** What's not done is the part that turns these CSVs into a live service — the FastAPI endpoints and the dashboard — which is why I asked what you want to build next.